

Hand in solutions of 7 from 8 problems below if you follow any of the 7.5hp courses (FMAN71, MATC70 and FMA122F) or 6 problems if you follow any of the 6hp courses (FMAN70, FMA121F). For a passing grade (3.5 for 7.5 points course and 3.0 for 6 points course), at least one problem of the last four have to be solved. Credits can be given for partially solved problems. Write your solutions (in English or Swedish) neatly and explain your calculations. Both the content and the format of your solutions, and also how difficult problems you choose, will affect your grade.

The exam should be submitted through the canvas page by January 15. **Write your name, section-year, and the code for the course you are following on the first page. If you have preferences for when to take the oral exam, also write this on the first page.** The solutions should be submitted as a pdf-file. We encourage you to hand in scanned hand-written solutions, but make sure that the solutions are easily readable after scanning. Programs that you have written (e.g. matlab or maple files) should also be submitted through the canvas page. Make sure to reference these and explain how you have used them in your main solution-pdf.

You may use books and computer programs (e.g. Matlab and Maple), but it is not permitted to get help from other persons (or chat bots). You do not need to explain the elementary matrix operation such as matrix multiplications made by the computer, but should explain steps in more complicated calculations such as jordanization. Ask if you are not sure!

You can use any theorem in the book without proof but not exercises without explaining the solutions.

All matrices below are complex matrices if nothing else is specified explicitly.

Good luck!

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1. Let V be the set of 2×2 matrices where the elements sum to zero, that is, $A = (a_{ij}) \in V$ if $\sum_{ij} a_{ij} = 0$. Show that V is a subspace of $M_2(\mathbb{C})$. If

$$M = \begin{pmatrix} 2 & 1 \\ -3 & -2 \end{pmatrix} \quad \text{and} \quad N = \begin{pmatrix} 2 & 1 \\ -3 & 4 \end{pmatrix}$$

determine which of the maps $\mathcal{M}(A) = MA$ and $\mathcal{N}(A) = NA$ are linear operators on V . Pick a basis for V and compute the matrix representations of maps above that are linear operators $V \rightarrow V$. Determine their characteristic and minimal polynomials.

2. Let A be a square matrix. Suppose that g is defined on A , f is defined on $g(A)$ and $f \circ g$, (where the composition is given by $f \circ g(x) = f(g(x))$), is defined on A . Show that $f \circ g(A) = f(g(A))$. Let

$$B = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix}.$$

Compute $C = e^B$ and verify that $\ln(C) = B$.

3. Show that if A is invertible and A^2 diagonalizable then A is also diagonalizable. Do there exist non-invertible matrices B such that B^2 is diagonalizable but not B ?
4. Let S be a 3×3 real skew-symmetric matrix and R a 3×3 real rotation matrix (that is, orthogonal with determinant 1). Prove or disprove (by counter example) the following statements:
 1. S has eigenvalues $0, ia, -ia$, where a is some real number.
 2. R has eigenvalues $1, e^{i\phi}, e^{-i\phi}$, where ϕ is some real number.
 3. $(I + S)$ is invertible
 4. $(I + R)$ is invertible if R does not have the eigenvalue -1 .
 5. $(I - S)(I + S)^{-1}$ is a rotation matrix.
 6. $(I - R)(I + R)^{-1}$ is skew-symmetric if R does not have the eigenvalue -1 .
5. Let A be a positive definite Hermitian matrix and $B = C^H A C$, where C is a matrix of the same size as A . Show that
 1. B is Hermitian and positive semidefinite.
 2. Show that matrices B and C have the same rank.
 3. Show that B is positive definite if and only if $\dim(\text{Ker}(C)) = 0$.
6. Let $(\cdot, \cdot)$ be a scalar product and $\|\cdot\|$ a norm on a vectorspace V . If A is a complex $m \times n$ matrix show that

$$\|v\|_* = \max_{\|u\| \leq 1} |(Au, Av)|$$

is a norm on V if $\text{Ker}(A) = \{0\}$.

7. Let A, B be square matrices of sizes $(m \times m)$ and $(n \times n)$ respectively, and $p_B(x)$ the characteristic polynomial of B . Show that the matrix $p_B(A)$ is invertible if and only if A and B have no common eigenvalues. Now consider the linear system of equations

$$AX - XB = C,$$

where C is a given matrix and X is unknown. How many unknowns and equations are there? Show that this system has precisely one solution when A and B have no common eigenvalues.

8. Let $W = (w_{ij})$ be a square symmetric matrix with real non-negative elements w_{ij} , and $w_{ii} = 0$. Furthermore, define a diagonal matrix D by $d_{ii} = \sum_j w_{ij}$ for each i . Show that the quadratic form

$$f(X) = X^T(D - W)X$$

is positive semidefinite and that $(D - W)$ has an eigenvalue that is 0. Suppose that X is an eigenvector with eigenvalue 0 for which $x_k \neq x_l$ for some choice of k, l . Show that then there is a binary vector Y (that is, with elements $y_i \in \{0, 1\}$) such that $f(Y) = 0$ and $y_k \neq y_l$.